Linguistic and Narrative Differences Across News Portals on Selected Topics

Dorjan Štrbac and Fran Kufrin

University of Zagreb, Faculty of Electrical Engineering and Computing, Zagreb, Croatia

Abstract

Purpose – This project studies whether different news portals show regular linguistic and narrative-related patterns when they report on selected topics. We use outlet classification as a measurable way to test this: if a model can identify the portal from the cleaned article body, the text contains repeatable portal-related differences. We also test the models on the other topic as an extra check of how topic-dependent those patterns are.

Design/Methodology/Approach – We built and cleaned an original corpus of 2,764 English-language articles from Al Jazeera, BBC and The Guardian. We used two topics: Israel/Gaza/Palestine conflict reporting (Topic A) and economy/finance/markets/trade reporting (Topic B). After controlling for publication month and article length, 2,567 articles were eligible for the experiment and were split into training, validation and sealed test sets. We compared direct TF-IDF classifiers, TF-IDF/SVD neural and classical models, and RoBERTa-base with either a frozen or fully fine-tuned encoder. For long articles, we compared the first 512-token chunk, a token-count-weighted mean of all chunks, and learned attention pooling over all chunks. The main metric was balance-weighted macro-F1.

Findings – The cleaned article bodies contain clear portal-predictive information on both topics. Direct TF-IDF already performs well in the same-topic tests (0.8741 on A→A and 0.8824 on B→B), while fully fine-tuned RoBERTa gives the strongest overall results (0.9215 for FT-ATTN on A→A and 0.9564 for FT-MEAN on B→B). The TF-IDF-based baselines perform well, but using SVD instead of the original TF-IDF representation or using an MLP instead of a classical classifier does not consistently improve performance. Full fine-tuning is much stronger than keeping RoBERTa frozen, and the additional test on the other topic shows a larger gap between lexical and fine-tuned contextual models. Attention pooling helps in several cases but is not always better than mean pooling, and its fine-tuned chunk weights remain close to the token-count baseline.

Originality/Value – The project uses a controlled outlet-classification experiment to study linguistic and narrative-related differences between three news portals on two topics. Careful cleaning and balancing reduce simple shortcuts, while the model comparison shows how lexical and contextual representations use the remaining differences. The second-topic test, attention analysis and PCA give extra information about how stable and structured the learned portal signal is, without assuming that any single model cue is a direct measure of narrative or style.

Keywords – news portals; linguistic patterns; narrative and framing; outlet classification; TF-IDF; RoBERTa; attention pooling.

Paper Type – Research paper.

1. Introduction

News articles are not only collections of facts about an event. They also contain repeated choices made by an outlet: the words it uses, how sentences are written, how sources are quoted or introduced, how an article is organized, and how different facts are connected. An outlet can also develop a recurring angle or broader narrative around a topic, for example by giving more attention to certain context, actors, causes or consequences. These choices can create patterns that are useful for recognizing the portal. At the same time, a classifier can become too easy to

interpret if it simply relies on an outlet name, author credit, repeated template, article length, or another obvious shortcut.

Our main goal is to study whether these portal-related differences are strong enough to be detected from article bodies after obvious source clues are reduced. Instead of manually assigning every article a linguistic or narrative label, we use outlet classification as a practical test: if a model can repeatedly identify Al Jazeera, BBC or The Guardian, then there are systematic differences in the text that the model can use. We study this on two selected topics. Topic A contains reporting related to Israel, Gaza and Palestine and the connected regional conflict. Topic B contains reporting about the economy, finance, markets, fiscal and monetary policy, and trade.

The two topics let us look at the same basic question in two different settings. A model trained on one topic is expected to use some topic-specific information too. That is not automatically a problem: repeated framing, background information or narrative choices inside a topic can be part of the real difference between portals. We therefore first evaluate normal same-topic classification (A→A and B→B). As an additional test, we also apply each selected model directly to the other topic (A→B and B→A) to see how much of the learned portal signal remains useful when the subject changes.

To study this, we compare TF-IDF representations, lower-dimensional SVD representations, small neural classifiers and pretrained Transformer models. RoBERTa is tested with a frozen encoder (FE) and with full fine-tuning (FT). Most articles are longer than one 512-token Transformer input, so we also compare three ways of building one article representation: only the first chunk (512), a token-count-weighted mean of all chunks (MEAN), and learned attention over all chunks (ATTN). Model selection uses validation data only, and the test set stays sealed until all predictive choices are fixed.

The questions we focus on are therefore broader than cross-topic transfer alone: can the portal be identified from the cleaned article body, which representations and model types work best, what changes between the two topics, and how much of the learned signal remains useful on the other topic? We also look at how long articles are combined from chunks and what attention and PCA can tell us about the learned representations without treating those analyses as direct explanations of every model decision.

2. Related Work

This project is related to stylometry and authorship attribution, where features of a text are used to identify its author or source. Surveys by Stamatatos (2009) and Koppel et al. (2009) show that useful clues can come from vocabulary, character patterns, syntax and document structure. Topic and genre can also get mixed together with these source-related patterns. Stamatatos (2017), for example, shows that authorship attribution becomes harder when the topic changes. For our project, the important lesson is that successful source classification can use several kinds of signals at once, so a high score does not by itself tell us exactly which type of difference the model found.

Research on news text also shows that publishers can differ in language, style and framing. Rashkin et al. (2017) found language differences among reliable news, satire, hoaxes and propaganda. Potthast et al. (2018) showed that writing style can help separate hyperpartisan and mainstream news, while Baly et al. (2018, 2020) studied factuality, bias and political ideology prediction in news. Card et al. (2015) also showed how news framing can be studied computationally by looking at which aspects of an issue are emphasized. This is close to the

broader idea behind our title: portal differences may appear not only in word choice, but also in recurring emphasis, the choice of sources and the way a topic is presented. Our classifier can benefit from such patterns, although it does not label them separately.

The second-topic test connects our work to domain shift. When training and test data come from different subjects, the distribution of words, names and events changes, and model performance can drop (Blitzer et al., 2007; Pan & Yang, 2010). This is one useful part of our experiment rather than its only goal. It is especially relevant for TF-IDF because its vocabulary and IDF values are learned from the source-topic training data, so cross-topic performance reflects the complete source-fitted pipeline, not only the final classifier.

Pretrained language models work differently. BERT showed the value of large-scale pretraining followed by fine-tuning for a specific task (Devlin et al., 2019), and RoBERTa improved the BERT training procedure (Liu et al., 2019). Gururangan et al. (2020) also showed that further adapting a language model to a domain or task can help. These models already have a subword vocabulary before our experiment and represent words in context, so they are less tied to the exact vocabulary of one source topic than TF-IDF is.

Long articles bring another practical problem because standard BERT/RoBERTa inputs are limited to 512 tokens. Hierarchical attention methods show that a document can be built from smaller parts while learning how much weight to give each part (Yang et al., 2016). However, attention weights should not automatically be treated as an explanation of a model decision. Jain and Wallace (2019) questioned how closely attention matches feature importance, while Wiegreffe and Pinter (2019) argued that the answer depends on how attention is used and interpreted. For this reason, we mainly use our chunk-attention weights to describe how the pooling layer behaves, not as a direct explanation of why RoBERTa predicted a certain outlet.

Taken together, this earlier work gives us a useful starting point. Source-related language and style patterns can be learned from text, framing research shows that news stories can differ in what they emphasize, TF-IDF gives a strong lexical baseline, and pretrained Transformers provide richer contextual representations. We therefore compare several model types on two topics, while using the cross-topic directions as an extra test of how stable the learned portal-related patterns are.

3. Methodology

3.1 Dataset Description

We collected articles from Al Jazeera, BBC and The Guardian from 28 July 2024 to 27 January 2026. We kept written news articles and removed page types such as live trackers, videos, opinion pieces, newsletters and digests, because these formats could create obvious structural differences. Topic A covers Israel/Palestine/Gaza reporting and directly related regional conflict, with the topic clearly identifiable from the article headline. Topic B covers the economy, finance, markets, fiscal or monetary policy, and trade. Candidate Topic B labels were checked again using the headline and article lead.

We started with 23,117 raw records. After checking whether pages could be fetched correctly, filtering by date and page type, checking genre and extraction quality, filtering by topic, removing source-identity clues, requiring at least 150 cleaned body words, and removing exact duplicates, 2,801 articles remained. Near-duplicate removal excluded another 37 articles, leaving a frozen corpus of 2,764. The near-duplicate removal rule was deliberately strict,

because we preferred to remove borderline cases rather than risk keeping very similar versions of the same article in different data splits.

Table 1 shows both the full frozen corpus and the smaller subset that remained eligible after the month/length controls described below. We did not force the raw corpus to contain exactly the same number of articles from every outlet. Instead, the experiment balances how much each outlet contributes inside publication-time and length groups, while fixed evaluation weights let us keep all eligible validation and test articles.

Table 1: Frozen corpus and experiment-eligible article counts by outlet and topic.

Outlet	Topic A frozen	Topic A eligible	Topic B frozen	Topic B eligible
Al Jazeera	599	558	197	188
BBC	353	346	360	325
The Guardian	470	460	785	690
Total	**1,422**	**1,364**	**1,342**	**1,203**

3.2 Dataset Preprocessing and Leakage Control

The classifiers use only the cleaned article body. Headlines are kept as metadata and for separate checks, but they are not used as input for the reported models. The goal of cleaning was to remove easy source clues without rewriting the normal article language. We remove or neutralize explicit outlet names, structural bylines and author credits, repeated source/program attributions, URLs, social-media handles, boilerplate and extraction artifacts. The main cleaning function is also label-blind: it does not know which outlet an article belongs to while cleaning it.

Exact and near-duplicate checks were done before splitting the data. This matters because having the same or almost the same story on both sides of a split could make the results look better than they really are. We also checked for remaining source identifiers and suspicious word patterns. These checks cannot remove every possible correlation, but they reduce the most obvious kinds of leakage.

Publication month and article length were also treated as possible shortcuts. For each topic and month, we created common length groups using portal-equal-weighted quantiles. The procedure first tried three length groups, then two, then one. A month was excluded if even one group could not provide at least seven articles from every outlet. This left 2,567 articles eligible for the experiment. We then used a 60% training, 20% validation and 20% test split inside each topic/month/length-bin/portal group. Topic A had 811 training, 272 validation and 281 test articles; Topic B had 721, 245 and 237.

Training and evaluation use this balancing in slightly different ways. Neural models use a deterministic sampler that keeps the outlets balanced inside each month-by-length group and then shuffles the samples globally each epoch. Validation and test sets are not downsampled. Instead, every eligible article gets a fixed evaluation weight so that the three outlets contribute equally inside the accepted groups. This gives effective balanced test masses of 186 articles for Topic A and 111 for Topic B while still evaluating all 281 and 237 actual test articles.

We also checked whether article length was still an easy shortcut after this balancing. A classifier using only log article length reached weighted validation accuracy of 0.334 on Topic A and 0.354 on Topic B, which is very close to the 0.333 chance level of a balanced three-class task. This does not prove that every structural difference is gone, but it suggests that article length by itself is no longer useful for guessing the outlet.

3.3 Theoretical Background: Text Representations and Models

Direct TF-IDF. Our main word-based baseline represents each article with word TF-IDF features (Salton & Buckley, 1988). The final search included unigrams and unigram-plus-bigram features, several minimum document-frequency values, and standard or sublinear term frequency. Eq. 1 shows a simplified form of the weighting before the final L2 normalization. Here N is the number of source-topic training documents and $df(t)$ is the number of training documents containing term t; the tf part is either the normal count or its sublinear version, depending on the configuration. One detail becomes important later: the TF-IDF vocabulary and IDF values are fitted on all source-topic training articles without balance weights. The later balancing therefore does not decide which terms enter the vocabulary or how their IDF values are calculated. The classical classifiers then use the training balance weights. We tested six classical families, including logistic regression, LinearSVC and linear, RBF, polynomial and sigmoid SVC variants (Cortes & Vapnik, 1995). LinearSVC was selected on validation data for both topics.

$$\mathrm{tfidf}(t,d) = \mathrm{tf}(t,d)\left[\ln\left(\frac{1+N}{1+\mathrm{df}(t)}\right)+1\right] \tag{1}$$

TF-IDF MLP and weighted SVD. We also trained small feed-forward neural networks directly on TF-IDF features. To create a smaller dense representation, we reduced TF-IDF with TruncatedSVD, which is related to latent semantic analysis (Deerwester et al., 1990). For each source topic and TF-IDF representation, SVD was fitted at 64, 128, 256 and 512 dimensions. To make the SVD fit follow the same balancing goal, each training row was multiplied by the square root of its balance weight, as shown in Eq. 2. The square root is used because SVD minimizes squared reconstruction error. After fitting, articles are transformed normally. We then tested both classical classifiers and MLPs on these SVD features. The MLP search used small one- and two-hidden-layer networks, dropout and a limited set of learning-rate and weight-decay choices, followed by five-seed confirmation for the shortlisted setups.

$$X_w = D_w^{1/2}X, \qquad D_w = \mathrm{diag}(w_1,\ldots,w_n) \tag{2}$$

RoBERTa. The Transformer experiments use the RoBERTa-base model and tokenizer (Liu et al., 2019). We compare two main training setups. With a frozen encoder (FE), RoBERTa is used as a fixed feature extractor and only the document pooling/classification head is trained. With full fine-tuning (FT), both the pretrained encoder and the head are trained for outlet classification. Selected FE setups were checked with five fresh seeds and selected FT setups with three fresh paired seeds. We also ran predefined random-initialized RoBERTa controls for all three pooling strategies on both source topics, using the pretrained tokenizer but random encoder weights. Each control used one seed, so these runs are only sanity checks, not fully tuned Transformers trained from scratch.

Most articles are longer than one Transformer input, so we split them into non-overlapping chunks of up to 510 content tokens plus RoBERTa special tokens. There is no article-level chunk limit for MEAN or ATTN, so these models can use all chunks. Each chunk gives a 768-dimensional RoBERTa vector, and we compare three ways of turning those chunk vectors into one article vector. 512 keeps only the first chunk. MEAN uses the token-count-weighted

average in Eq. 3, where h_i is the vector for chunk i and n_i is its number of content tokens. ATTN uses a small learned scorer (768→128→1). Eq. 4 shows how the scores become normalized chunk weights while still using log chunk length as a starting preference, and Eq. 5 gives the final article vector. The scorer does not use an explicit chunk-position embedding and there is no Transformer layer between chunks. Its final score layer starts at zero, which means that at the beginning α_i is exactly proportional to n_i and ATTN starts from the same weighting as MEAN. The final 768-dimensional article vector then goes through dropout and a 768→3 linear classifier.

$$h_{\text{MEAN}} = \frac{\sum_{i=1}^{m} n_i h_i}{\sum_{i=1}^{m} n_i} \tag{3}$$

$$e_i = v^{\top}\tanh(Wh_i + b) + \ln n_i, \qquad \alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^{m}\exp(e_j)} \tag{4}$$

$$h_{\text{ATTN}} = \sum_{i=1}^{m} \alpha_i \, h_i \tag{5}$$

3.4 Model Selection and Evaluation Protocol

We made all predictive choices before opening the final test set. Training data was used to fit model parameters, validation data was used to choose representations, settings and checkpoints, and test article text was not loaded or processed during model selection. Only after all model and representation choices for TF-IDF, SVD, MLP and RoBERTa were finalized did we open the test set in a separate final-evaluation notebook.

Our main metric is balance-weighted macro-F1. We calculate precision and recall for each outlet using the fixed article weights, convert them to an F1 score, and then average the three outlet F1 scores equally, as shown in Eq. 6. This keeps every outlet equally important while the article weights keep the month/length balancing used for validation and test evaluation. We also saved weighted accuracy and per-class precision, recall and F1 as extra metrics. For neural setups trained with several seeds, the main final number is the mean test weighted macro-F1 across the predefined confirmation seeds, with sample standard deviation showing how much the result changed between runs. We also average seed probabilities to create a secondary ensemble, but the main comparison remains the mean across individual seeds.

$$F_{1,\text{macro}}^{(w)} = \frac{1}{3}\sum_{c=1}^{3} \frac{2P_c^{(w)}R_c^{(w)}}{P_c^{(w)}+R_c^{(w)}} \tag{6}$$

The searches were meant to cover a useful range of settings while still being realistic to run. They were not meant to try every possible model and hyperparameter combination. The classical TF-IDF grid was fairly broad, while the neural and Transformer searches were narrower or done in stages because they were much more expensive. The random-initialized Transformer was only a predefined control, not a fully tuned competing model. We return to this limitation later.

The model-selection stage was deliberately broader than the final headline table. The direct classical stage compared 16 TF-IDF representations across six classifier families. The TF-IDF/SVD stage then tested direct MLPs, four weighted SVD sizes (64, 128, 256 and 512), the same classical families on SVD features, and staged MLP searches on the reduced representations. The Transformer stage compared 512, MEAN and ATTN under both frozen-encoder and full-fine-tuning setups, followed by fresh multi-seed confirmation. This lets us compare not only one strong model against another, but also what changes when we move from sparse lexical features to compressed dense features and then to pretrained contextual representations.

4. Results

4.1 Main Classification Results

Table 2 summarizes the main test results. The first important point is that the portal can be predicted quite well from the cleaned article body on both topics. Direct classical TF-IDF reaches weighted macro-F1 of 0.8741 on A→A and 0.8824 on B→B. The fine-tuned RoBERTa variants are stronger still: FT-ATTN reaches 0.9215 on A→A, while FT-MEAN reaches 0.9564 on B→B. This suggests that both lexical patterns and richer contextual information carry useful differences between the three portals.

The cross-topic directions add a second view of the same models. Direct TF-IDF drops to 0.5738 on A→B and 0.4599 on B→A, while fine-tuned RoBERTa keeps much more of its performance. FT-ATTN reaches 0.8863 on A→B and 0.7607 on B→A. Attention is not automatically the best pooling method: on B→B, FT-MEAN has the highest main mean-seed score (0.9564), compared with 0.9448 for FT-ATTN. So the main picture is not simply that one architecture wins everywhere; several fine-tuned variants are strong, with somewhat different behavior across the two topics.

Table 2: Primary test weighted macro-F1. Neural values are mean ± sample SD across predefined confirmation seeds.

Direction	Direct classical	FT-512	FT-MEAN	FT-ATTN
A→A	0.8741	0.9083 ± 0.0219	0.9133 ± 0.0284	**0.9215 ± 0.0104**
A→B	0.5738	0.8662 ± 0.0086	0.8766 ± 0.0276	**0.8863 ± 0.0483**
B→A	0.4599	0.7055 ± 0.0450	0.7166 ± 0.0242	**0.7607 ± 0.0274**
B→B	0.8824	0.9278 ± 0.0339	**0.9564 ± 0.0196**	0.9448 ± 0.0178

Figure 1 gives a broader view of the Topic A same-topic test and includes all of the main configuration families. It shows that the task is already meaningful before we ask about topic changes: direct TF-IDF, direct MLP and SVD-based models all classify the portal reasonably well, while the fully fine-tuned RoBERTa variants give the strongest results. Frozen RoBERTa and the random-initialized controls are much weaker. This wider comparison is useful because it shows that the final result is not based on only one baseline and one Transformer model.

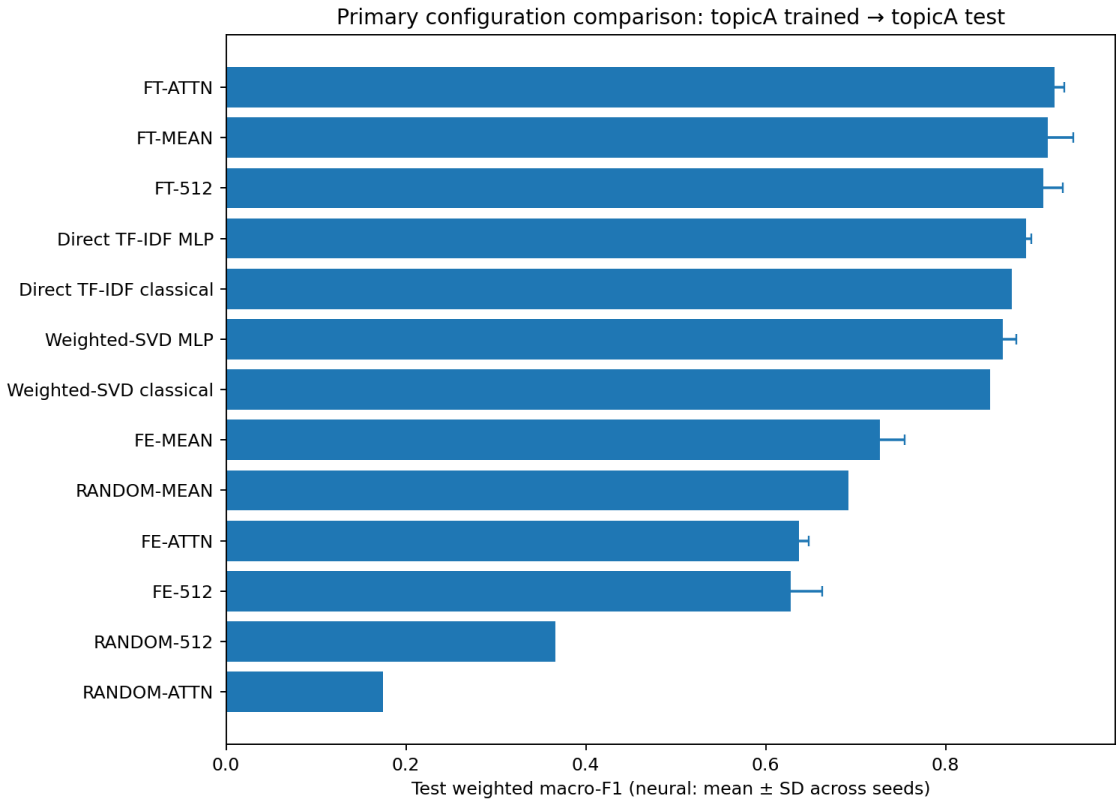


Figure 1: Full primary configuration comparison for Topic A training and Topic A testing. Error bars show seed SD for multi-seed neural configurations.

4.2 Direct TF-IDF, SVD and MLP Baselines

The non-Transformer experiments also contain some useful results that are easy to lose if we look only at the final RoBERTa scores. Table 3 compares the four main TF-IDF/SVD pipelines. On Topic A, direct TF-IDF is slightly better than the weighted-SVD versions for both the classical and MLP classifiers. Topic B is different: weighted SVD improves the classical result slightly and gives a much larger gain for the MLP (0.8770 instead of 0.8004 on B→B). The same pattern is visible on B→A, where SVD also helps both baseline families. This means that dimensionality reduction was not simply better or worse; its effect depended on the topic and the classifier.

The direct MLP is also not automatically better just because it is neural. It performs well on A→A, but the selected LinearSVC is stronger on Topic B and transfers better in both directions. According to these results, the way the text is represented — direct TF-IDF or the reduced SVD representation — can affect performance just as much as the choice of classifier.

Table 3: Test weighted macro-F1 for the main TF-IDF and weighted-SVD baseline families.

Pipeline	**A→A**	**A→B**	**B→A**	**B→B**
Direct classical	0.8741	0.5738	0.4599	0.8824
Direct MLP	0.8893 ± 0.0064	0.4940 ± 0.0267	0.3609 ± 0.0247	0.8004 ± 0.0158
SVD classical	0.8497	0.5556	0.5094	0.8913
SVD MLP	0.8639 ± 0.0143	0.5161 ± 0.0221	0.4445 ± 0.0566	0.8770 ± 0.0142

4.3 What Pretraining and Fine-Tuning Add

The RoBERTa experiments were designed to separate three related questions: what the architecture can do with randomly initialized encoder weights, what is available from a pretrained encoder used only as a fixed feature extractor, and what happens when the pretrained encoder is allowed to adapt to our task. Figure 2 shows this comparison on A→A for all three document-pooling strategies. The jump from FE to FT is large for 512, MEAN and ATTN. The random-initialized controls are useful as sanity checks, but each used only one predefined setup and one seed, so they should not be read as optimized "training from scratch" results.

The matched comparison also shows why the 512/MEAN/ATTN choice should not be discussed separately from encoder training. With a frozen encoder, MEAN is clearly the strongest of the three on A→A. After full fine-tuning, all three become strong and much closer together, with ATTN slightly ahead in the primary mean-seed result. This suggests that adapting the contextual encoder to the outlet task matters more than choosing a complicated pooling method by itself.

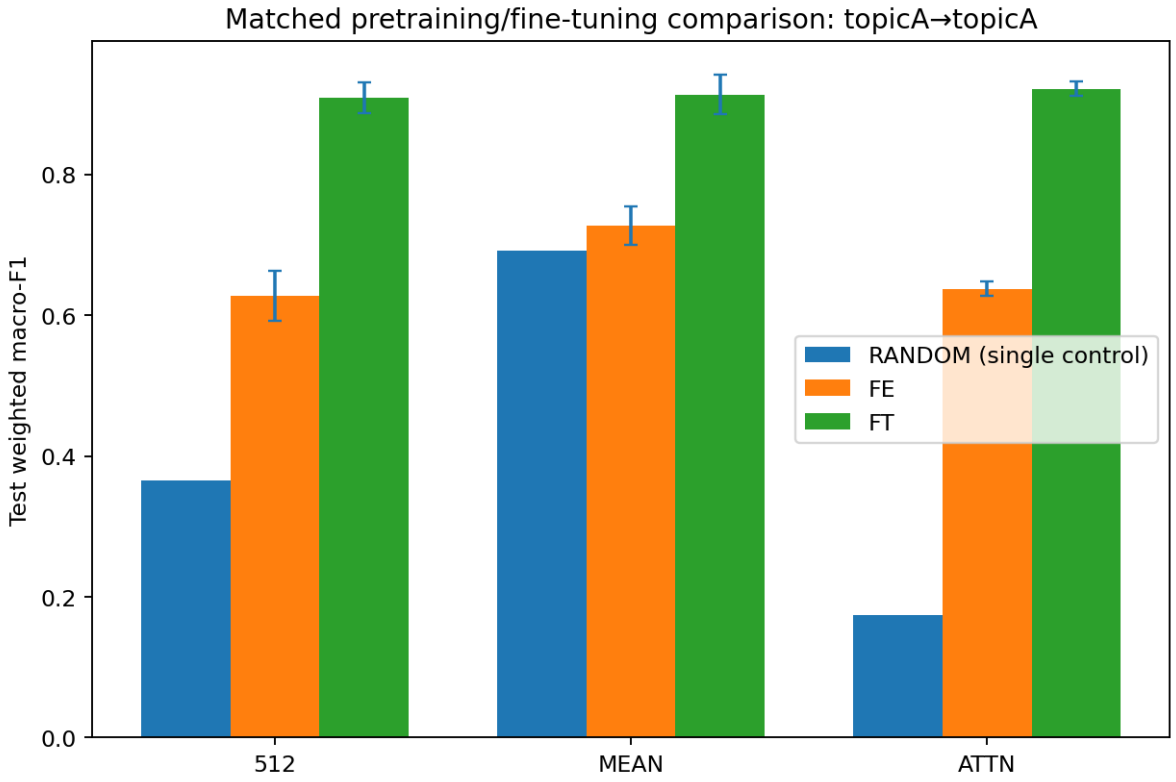


Figure 2: Matched comparison of random-initialized, frozen-encoder (FE) and fully fine-tuned (FT) RoBERTa on A→A. Error bars show seed SD where several confirmation seeds were used.

4.4 How Results Change Across Topics

After the same-topic tests, we also asked what happens when a selected model is applied to the other topic without adapting it. This is an extra stress test, not an expectation that a model trained on one topic should ignore everything specific to that topic. The two directions are clearly not equally difficult. The direct classical pipeline keeps about 65.7% of its within-topic score on A→B, but only about 52.1% on B→A. FT-ATTN keeps about 96.2% on A→B and about 80.5% on B→A.

This asymmetry is interesting, but it does not mean that Topic A is simply a 'more general' topic. The two corpora differ in events, vocabulary, outlet coverage, class distributions, and the framing or narratives that may appear inside each topic. Several of these things can affect how well a model carries over. The safer conclusion is simply that, in this experiment, models trained on Topic A transfer to Topic B more successfully than the other way around.

Per-class scores also show that errors are not distributed evenly across the three portals. For example, on A→B the selected direct classical model has very low BBC recall (about 0.142 after balancing), while the secondary FT-ATTN probability ensemble raises it to about 0.748. We use this as an error-analysis example rather than as the headline result; the main architecture comparison remains the mean across individual confirmation seeds in Table 2.

As an extra check, we also used paired stratified bootstrap resampling. For the FT-ATTN ensemble minus direct classical, the 95% intervals for the weighted macro-F1 difference are positive in all four directions: [0.013, 0.099] on A→A, [0.275, 0.382] on A→B, [0.290, 0.390] on B→A and [0.037, 0.115] on B→B. FT-ATTN and FT-MEAN are much closer to each other; only B→A shows a clearly positive ensemble difference. This supports the broader result that fine-tuned RoBERTa is strong overall, while ATTN and MEAN remain fairly close alternatives.

4.5 Error Patterns and Prediction Confidence

Macro-F1 tells us how well the three classes are handled overall, but the confusion matrices show what kind of mistakes are being made. Figure 3 compares the selected direct classical pipeline with the secondary FT-ATTN soft-probability ensemble on A→B. The lexical model predicts The Guardian very often and recognizes only a small part of the balanced BBC mass correctly. The FT-ATTN ensemble produces a much more diagonal matrix and raises balanced BBC recall from about 0.142 to about 0.748. We use the ensemble here only to make the error pattern easy to see; the main architecture numbers in Table 2 remain the mean across individual confirmation seeds.

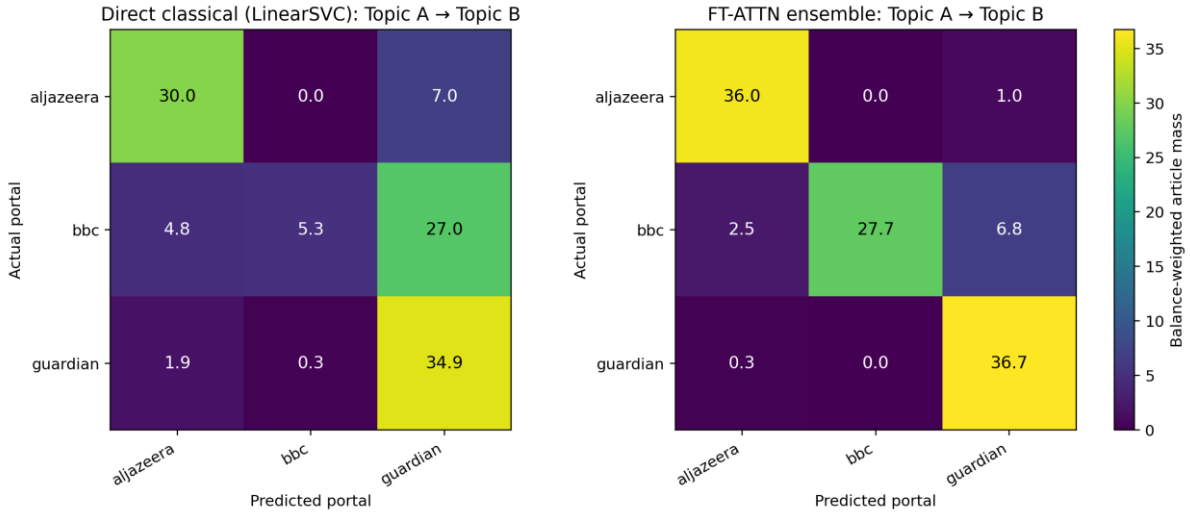


Figure 3: Balance-weighted confusion matrices for A→B. Left: selected direct TF-IDF classical model. Right: secondary FT-ATTN probability ensemble.

We also looked at the softmax confidence of the FT-ATTN ensemble after all predictive choices were fixed. In every direction, correct predictions are more confident on average than wrong predictions. For example, on A→A the weighted mean maximum softmax probability is

about 0.969 for correct predictions and 0.773 for wrong ones; on B→A it is about 0.892 versus 0.746. Figure 4 shows the same pattern across all four directions. These probabilities are not calibrated confidence estimates, so this is only a descriptive check, but it is still useful to see that the model is generally less certain when it makes a mistake.

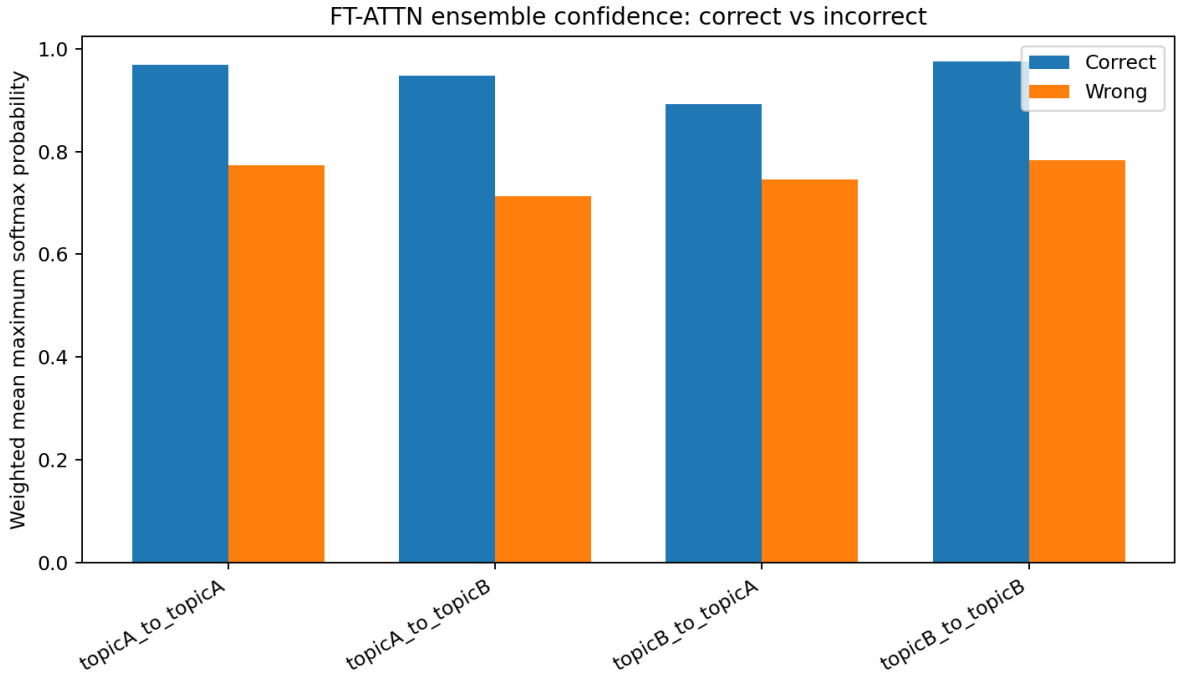


Figure 4: FT-ATTN ensemble weighted mean maximum softmax probability for correct and incorrect predictions in all four evaluation directions.

4.6 How Attention Behaves on Long Articles

Most test articles need more than one Transformer chunk: 94.4% of the balance-weighted Topic A test mass and 89.6% of Topic B are multi-chunk. So the way chunks are combined matters for most of the dataset, not only for a few very long articles. To compare ATTN with the token-count-weighted MEAN starting point, we use article-level total-variation (TV) distance, defined in Eq. 7. Here α_i is the learned ATTN weight and $p_i = \frac{n_i}{\sum_j n_j}$ is the MEAN/token-count baseline. A TV value near zero means the learned weights stayed very close to that baseline; a larger value means the model moved more weight from some chunks to others.

$$\mathrm{TV}(\alpha, p) = \frac{1}{2}\sum_{i=1}^{m}|\alpha_i - p_i|, \qquad p_i = \frac{n_i}{\sum_{j=1}^{m} n_j} \tag{7}$$

Frozen and fine-tuned models behave very differently here. For models trained on Topic B, FE-ATTN has weighted mean TV distances of about 0.254 on A and 0.228 on B. FT-ATTN stays much closer to the baseline, at about 0.015 and 0.009. Figure 5 shows this difference. The small FT changes do not mean that attention was useless. ATTN is still a separate trainable pooling method with a different training path, and even small but consistent changes in chunk weights can change the final article representation. At the same time, the B→B result in Table 2 reminds us that these changes are not always better than MEAN.

Our attention pooling is not aware of chunk order. It does not know whether a chunk comes first, second, or later in the article. Its weight for a chunk is therefore based on information from the chunk itself rather than its position. If earlier or later chunks receive different weights, this can only happen indirectly because their content, and hence the information available to the pooling layer, is different.

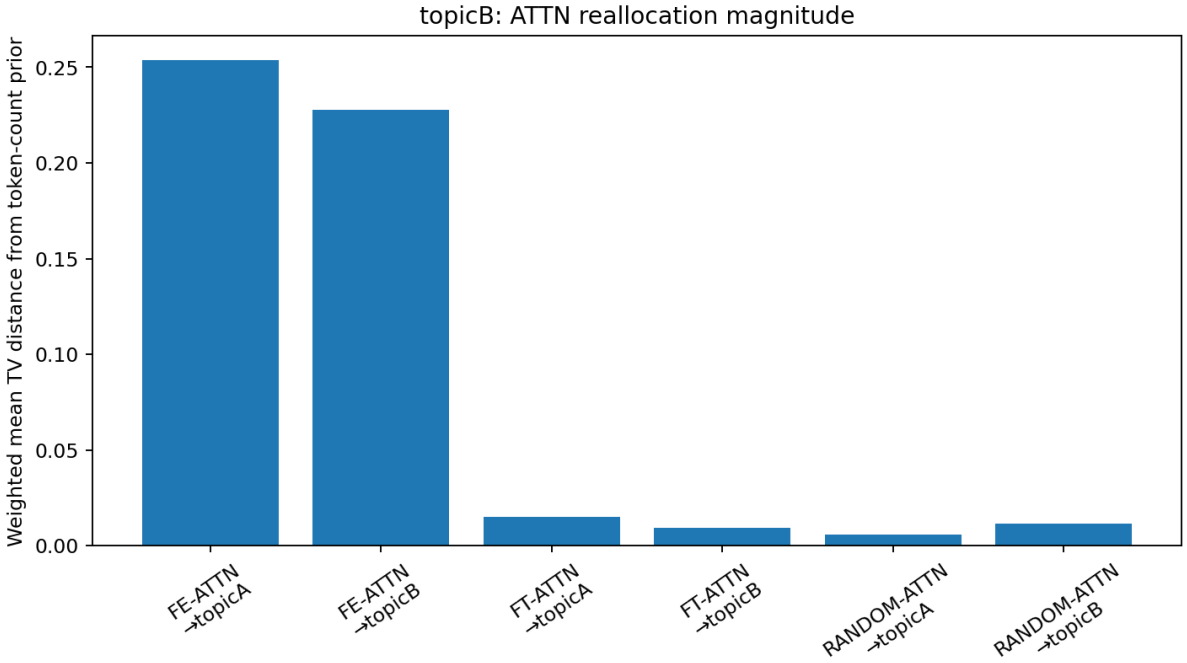


Figure 5: Magnitude of learned ATTN reallocation from the token-count prior for models trained on Topic B. Larger TV distance means stronger redistribution of chunk weight.

4.7 Linguistic Cues and the Learned Representation

The selected direct classical model is LinearSVC, so its coefficients give us a simple way to look at some of the lexical cues that are associated with each outlet. Table 4 lists a small set of readable high-positive features from the models trained separately on Topic A and Topic B. These are not the most common words in each portal and they do not prove a specific narrative. They are simply features that push this particular linear classifier toward one outlet rather than the others. The table shows both clearly linguistic habits, such as the BBC's use of titles such as 'Mr' and 'Ms', and topic/coverage-related cues, such as economic terms or different names used for actors and events.

The most prominent lexical cues also change between Topic A and Topic B. Some words or expressions may be useful for recognizing an outlet in both topics, while many others are closely connected to the people, events and subtopics covered in that particular dataset.

Table 4: Examples of high-positive LinearSVC TF-IDF features for each outlet. Features are selected from the final models trained separately on Topic A and Topic B for readability.

Outlet	Topic A examples	Topic B examples
Al Jazeera	captives; enclave; United Nations	percent; billion; IMF
BBC	Hamas-run; Mr; IDF	Mr; Ms; Scotland
The Guardian	airstrikes; militants; spokesperson	Britain; Australia; RBA

After model selection was complete, we used principal component analysis (PCA) only as a descriptive way to examine the learned representations and gain some intuition about their structure. Figure 6 shows a joint-topic weighted PCA of the Topic-A-trained FT-ATTN article representation, fitted on the whole experiment-eligible corpus. Each point starts from the 768-dimensional whole-article vector after all chunks have already been combined by the learned ATTN pooling and before the final dropout/linear classifier. The first two PCA components explain about 80.8% of weighted variance (46.2% and 34.6%). The plot shows clear portal-related structure, while the two topics also overlap in parts of that structure. This gives us a visual example of how the learned representation organizes the articles, but it does not by itself tell us which linguistic or narrative features created that structure.

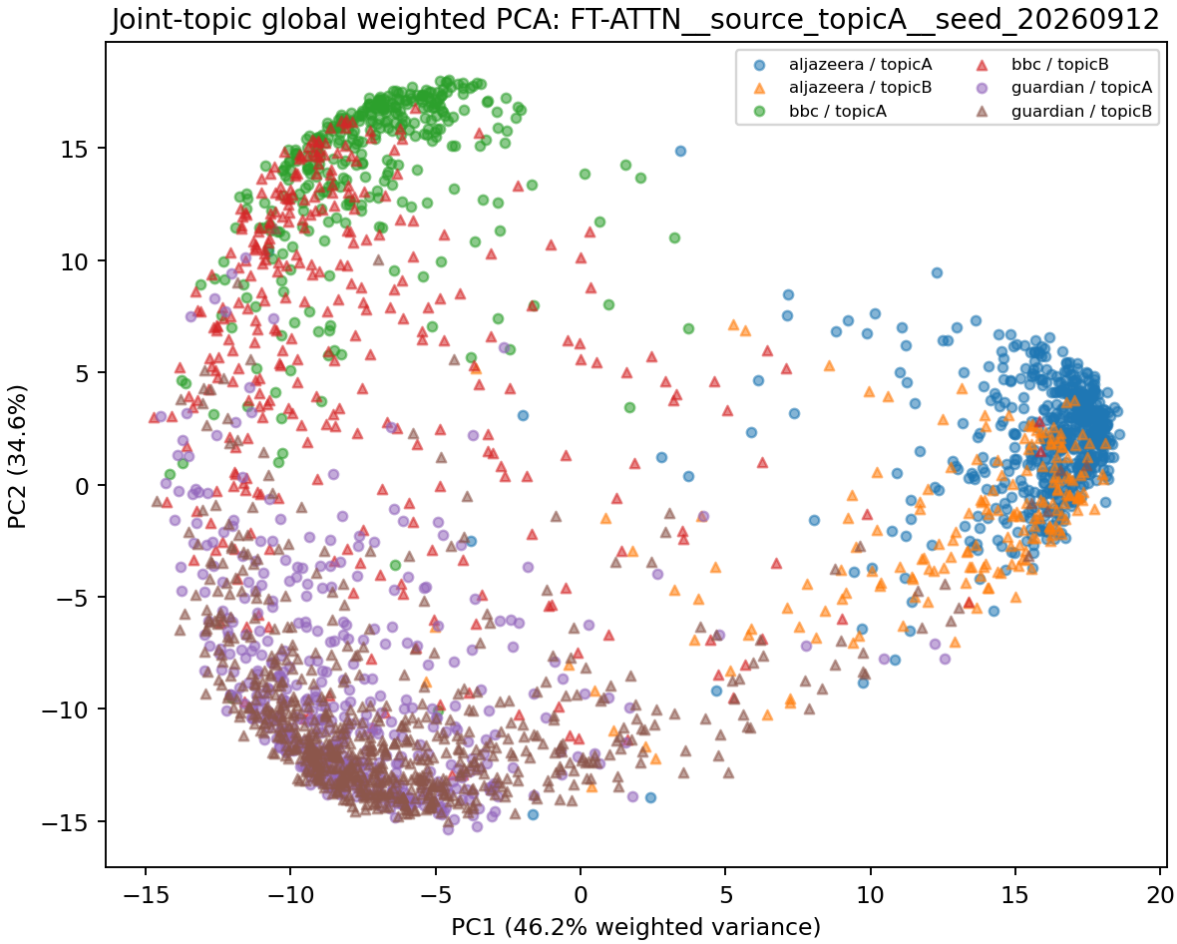


Figure 6: Joint-topic weighted PCA of the Topic-A-trained FT-ATTN article representation for one predefined seed, fitted on all 2,567 experiment-eligible articles from both topics and all three splits. Topic A and Topic B are normalized to equal total weight. Marker shape shows the topic, while each color in the legend represents one outlet-topic combination.

5. Discussion

5.1 Conclusions

The most basic result is that the three portals can be distinguished from their cleaned article bodies on both selected topics. Direct TF-IDF is already a strong baseline, which means that recurring word and phrase patterns contain useful portal information. Fine-tuned RoBERTa performs even better, suggesting that the contextual representations capture useful information beyond the lexical baseline. These results fit the original idea of the project: the portals show

regular differences in how they write and present the selected topics, and those differences are strong enough to be learned by several NLP models.

At the same time, classification accuracy does not tell us exactly which difference caused a prediction. A model may use vocabulary, sentence structure, the choice of sources and the way they are presented, repeated background information, framing, or parts of a broader narrative around the topic. Fine-tuned RoBERTa is the strongest overall approach we tested, both in the normal same-topic task and in the harder cross-topic test. The cross-topic results are useful because they show that some of the learned portal signal remains useful outside the training topic, but we do not expect a model trained on one topic to learn only topic-independent information.

5.2 Theoretical Implications

The cross-topic gap between TF-IDF and RoBERTa also needs to be read at the pipeline level. TF-IDF learns its vocabulary and IDF values only from the source-topic training texts. When we test it on the other topic, those articles are represented in a word space built from a different subject, and n-grams that appear only in the target topic cannot become new features. So part of the cross-topic drop can come from the representation itself before we even ask what the classifier learned. This does not reduce the value of TF-IDF as a strong same-topic baseline; it simply explains why its second-topic result should not be interpreted too narrowly.

Narrative is especially difficult to separate from the other signals. A portal may repeatedly emphasize certain background information, sources, actors, causes or consequences, or connect events in a characteristic way. TF-IDF can pick up repeated words and phrases through which such an angle is expressed, while RoBERTa can also represent more contextual versions of the same patterns. Our experiment does not label narratives or frames directly, so it cannot say how much of the classification score comes from narrative rather than vocabulary, style or structure. It does show that the article bodies contain a mixture of portal-related patterns that can be learned, and the cross-topic test gives one extra view of how much of that mixture remains useful when the topic changes.

The TF-IDF/SVD results show that a more compact representation is not automatically better. SVD helps the Topic B MLP substantially and also improves the B-trained classical model slightly, but direct TF-IDF is stronger for both baseline families on Topic A. One possible reading is that Topic B contains lexical variation that benefits more from a lower-dimensional latent representation, while Topic A keeps useful detail in the sparse features. We should keep this as an observation about these data rather than a general rule about SVD.

RoBERTa starts with a pretrained subword vocabulary and a contextual encoder, and full fine-tuning then adjusts the encoder for the outlet-classification task. Across the results, fully fine-tuned models are much stronger than the frozen versions, while the random-initialized control is weaker again. This suggests that both pretraining and task-specific fine-tuning matter. The advantage is visible in same-topic classification and becomes even more noticeable when the topic changes.

The coefficient examples give a more concrete view of what a lexical model can use. They include wording habits, preferred terminology, names of institutions or actors, and subject-specific vocabulary. These cues help explain why same-topic TF-IDF works so well, but they also show why classification is not the same as directly measuring narrative. A high coefficient can reflect language choice, coverage, geography, recurring sources, or several of these at once.

The attention results add another useful detail. Fine-tuned ATTN models can perform very well even when their final chunk weights stay close to the token-count baseline. In other words, a small change in the attention weights does not mean that the attention setup had no value. Frozen models move the weights much more, but their predictions are still worse. This matches the general warning from earlier work (Jain & Wallace, 2019; Wiegreffe & Pinter, 2019): attention weights can help us understand how the pooling layer behaves, but they should not be treated as a direct explanation of the whole classification decision.

The confidence analysis gives a small practical check on model behavior. Wrong FT-ATTN ensemble predictions are less confident on average than correct ones. That does not make the softmax probabilities calibrated, but it suggests that uncertainty information could be useful if a future system needed to flag borderline cases for manual review rather than always forcing a confident-looking label.

5.3 Practical Implications

For media analysis, the results show that portal-related patterns can be detected from article text even after obvious source clues are removed. At the same time, those patterns should always be interpreted in relation to the topic and dataset. If a future system is expected to work on different topics or future news events, it is useful to test that directly, instead of assuming that good results on a normal train/test split will stay the same in new situations.

The project also shows why it is useful to control simple shortcuts before trying to interpret the results. Removing obvious outlet names, preventing duplicate leakage, and controlling publication month and article length makes the task less dependent on clues that are not very interesting for our main question. The same problem can appear in many text-classification tasks when labels accidentally line up with formatting, time, document length or the way the data was collected.

Our results answer a fairly narrow question: how well these models can recognize which of the three portals an article came from. They do not by themselves tell us that one outlet is more accurate, less biased, or better journalism than another. A real media-analysis tool would also need to handle new outlets, changing editorial styles, other languages and cases where the model is too uncertain to make a useful guess.

5.4 Limitations and Future Research

This study covers only three English-language outlets, two topics and one publication period. The results therefore describe these portals and these topic distributions; we cannot assume that the same model ranking or the same linguistic patterns would appear for other publishers, languages, years or subjects. The corpus is also smaller than many large text-classification datasets. Its main strength is the amount of cleaning and checking rather than its size. A larger dataset with more outlets and topics would make it possible to test which differences are really stable and which are tied to one topic or period.

A second limitation is that we did not directly annotate linguistic features, frames or narratives. Classification gives us evidence that systematic portal-related differences exist, but it does not separate them into clean categories. Future work could compare articles about the same event more directly, annotate framing or narrative choices, or extract recurring themes and source patterns before comparing them across portals. It could also add more topics, test changes over time, use event-level splits, where all articles about the same event are kept in the same split so that test events are completely unseen during training, and evaluate the models on completely unseen outlets.

Our model search was useful, but it was not exhaustive. The classical TF-IDF stage used a fairly broad predefined grid, while the neural and Transformer stages used narrower or staged searches that were realistic for the available computing time. A larger study could try more learning rates, batch sizes, dropout values, optimizers, schedulers, seeds, layer-freezing strategies and document-pooling methods. The random-initialized RoBERTa controls each used only one predefined setup and one seed, so they should not be treated as strong representatives of what carefully tuned RoBERTa models trained from scratch could achieve. Some of these runs even collapsed during training. At the same time, training such a large model from scratch on our relatively small dataset and getting it to generalize well would be expected to be difficult.

There is also room to improve how long articles are handled. Our ATTN layer is a lightweight way of combining chunk vectors and does not use chunk order directly or let chunks interact with each other. Hierarchical Transformers, long-context models, position-aware pooling or methods that first select the most useful chunks could test whether a richer view of the whole article improves classification and makes the learned differences easier to study. If explanation were the main goal, the attention analysis could also be combined with tests that change or remove parts of the text and check how the prediction reacts.

Finally, the TF-IDF coefficients, PCA and confidence/entropy plots are extra analyses that we ran after the predictive setup was already fixed. They help us look at the models from different angles, but none of them is a direct explanation of the full decision process. A large TF-IDF coefficient is only an association inside one linear model trained on a particular topic, PCA compresses 768 dimensions into two, and softmax confidence shows how strongly the model favors one class, but it should not be treated as an exact probability that the prediction is correct. Future work could add more direct linguistic annotation and tests where parts of the text are changed or removed, so these observations can be connected more clearly to framing, narrative, the choice of sources, and the way sources are presented in the article.

6. References

Baly, R., Karadzhov, G., Alexandrov, D., Glass, J., & Nakov, P. (2018). Predicting factuality of reporting and bias of news media sources. In Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (pp. 3528-3539). Association for Computational Linguistics. https://doi.org/10.18653/v1/D18-1389

Baly, R., Da San Martino, G., Glass, J., & Nakov, P. (2020). We can detect your bias: Predicting the political ideology of news articles. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (pp. 4982-4991). Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.emnlp-main.404

Blitzer, J., Dredze, M., & Pereira, F. (2007). Biographies, Bollywood, boom-boxes and blenders: Domain adaptation for sentiment classification. In Proceedings of the 45th Annual Meeting of the Association of Computational Linguistics (pp. 440-447). Association for Computational Linguistics. https://aclanthology.org/P07-1056/

Card, D., Boydstun, A. E., Gross, J. H., Resnik, P., & Smith, N. A. (2015). The Media Frames Corpus: Annotations of Frames Across Issues. In Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Vol. 2, pp. 438-444). Association for Computational Linguistics. https://doi.org/10.3115/v1/P15-2072

Cortes, C., & Vapnik, V. (1995). Support-vector networks. Machine Learning, 20, 273-297. https://doi.org/10.1007/BF00994018

Deerwester, S., Dumais, S. T., Furnas, G. W., Landauer, T. K., & Harshman, R. (1990). Indexing by latent semantic analysis. Journal of the American Society for Information Science, 41(6), 391-407. https://doi.org/10.1002/(SICI)1097-4571(199009)41:6%3C391::AID-ASI1%3E3.0.CO;2-9

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional Transformers for language understanding. In Proceedings of NAACL-HLT 2019 (pp. 4171-4186). Association for Computational Linguistics. https://doi.org/10.18653/v1/N19-1423

Gururangan, S., Marasović, A., Swayamdipta, S., Lo, K., Beltagy, I., Downey, D., & Smith, N. A. (2020). Don't stop pretraining: Adapt language models to domains and tasks. In Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (pp. 8342-8360). Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.acl-main.740

Jain, S., & Wallace, B. C. (2019). Attention is not explanation. In Proceedings of NAACL-HLT 2019 (pp. 3543-3556). Association for Computational Linguistics. https://doi.org/10.18653/v1/N19-1357

Koppel, M., Schler, J., & Argamon, S. (2009). Computational methods in authorship attribution. Journal of the American Society for Information Science and Technology, 60(1), 9-26. https://doi.org/10.1002/asi.20961

Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., & Stoyanov, V. (2019). RoBERTa: A robustly optimized BERT pretraining approach. arXiv:1907.11692. https://doi.org/10.48550/arXiv.1907.11692

Pan, S. J., & Yang, Q. (2010). A survey on transfer learning. IEEE Transactions on Knowledge and Data Engineering, 22(10), 1345-1359. https://doi.org/10.1109/TKDE.2009.191

Potthast, M., Kiesel, J., Reinartz, K., Bevendorff, J., & Stein, B. (2018). A stylometric inquiry into hyperpartisan and fake news. In Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Vol. 1, pp. 231-240). Association for Computational Linguistics. https://doi.org/10.18653/v1/P18-1022

Rashkin, H., Choi, E., Jang, J. Y., Volkova, S., & Choi, Y. (2017). Truth of varying shades: Analyzing language in fake news and political fact-checking. In Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing (pp. 2931-2937). Association for Computational Linguistics. https://doi.org/10.18653/v1/D17-1317

Salton, G., & Buckley, C. (1988). Term-weighting approaches in automatic text retrieval. Information Processing & Management, 24(5), 513-523. https://doi.org/10.1016/0306-4573(88)90021-0

Stamatatos, E. (2009). A survey of modern authorship attribution methods. Journal of the American Society for Information Science and Technology, 60(3), 538-556. https://doi.org/10.1002/asi.21001

Stamatatos, E. (2017). Authorship attribution using text distortion. In Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics (Vol. 1, pp. 1138-1149). Association for Computational Linguistics. https://aclanthology.org/E17-1107/

Wiegreffe, S., & Pinter, Y. (2019). Attention is not not explanation. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and IJCNLP (pp. 11-20). Association for Computational Linguistics. https://doi.org/10.18653/v1/D19-1002

Yang, Z., Yang, D., Dyer, C., He, X., Smola, A., & Hovy, E. (2016). Hierarchical attention networks for document classification. In Proceedings of NAACL-HLT 2016 (pp. 1480-1489). Association for Computational Linguistics. https://doi.org/10.18653/v1/N16-1174